NPA – Exercise 1

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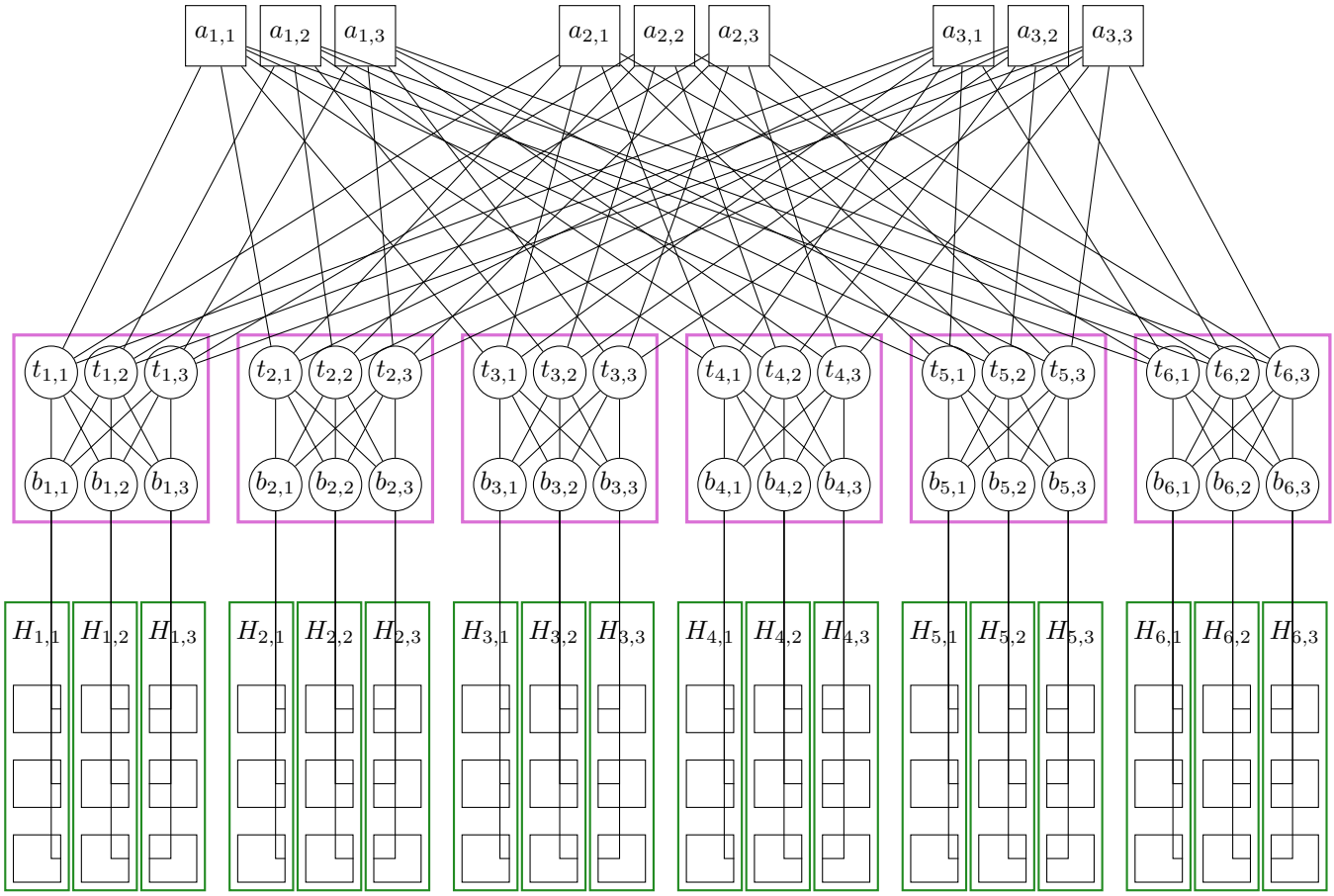
Exercise 1

(a)



(b)

As per definition, in each backbone group ℓ every switch $a_{\ell,j}$ should be connected to the j-th top-switch in each pod. Thus, $a_{1,2}$ should be connected to every $t_{i,2}$.

This is however not the case as we can see in Figure 1: $a_{1,2}$, labeled with `10.4.1.2`, is connected to $t_{1,1}$, labeled with `10.0.2.1`.

Only the second redundancy group is connected to the second top-switches. This does not fit the definition and thus, the topology displayed in Figure 1 is not a Clos topology.

(c)

The $\mathcal{C}(k)$ consists of

- $\frac{k}{2} \cdot \frac{k}{2}$ backbone-switches
- $k \cdot \frac{k}{2}$ top-switches
- $k \cdot \frac{k}{2}$ bottom-switches

Summing this up, the $\mathcal{C}(k)$ requires $\frac{5k^2}{4}$ switches.

For each of the $k \cdot \frac{k}{2}$ bottom-switches, there exist $\frac{k}{2}$ hosts, resulting in $k \cdot \frac{k}{2} \cdot \frac{k}{2} = \frac{k^3}{4}$ hosts.

Regarding the number of links, there are three layers.

First is the host-layer, where one link per host is needed, resulting in $\frac{k^3}{4}$ links.

Second there is the pod-layer, where in each of the k pods, each of the $\frac{k}{2}$ top-switches is connected to each of the $\frac{k}{2}$ bottom-switches. This leads to $k \cdot \frac{k}{2} \cdot \frac{k}{2} = \frac{k^3}{4}$ links for the pod layer.

Last, the backbone-switches have to be connected to the top-switches. For that, for each group of the $\frac{k}{2}$ backbone-switch groups, each of the $\frac{k}{2}$ backbone-switches in this group is connected to k top-switches, resulting in $\frac{k}{2} \cdot \frac{k}{2} \cdot k = \frac{k^3}{4}$ links on this layer.

Summing the three layers together, leads to a number of $\frac{k^3}{4} + \frac{k^3}{4} + \frac{k^3}{4} = \frac{3k^3}{4}$ necessary links for a $\mathcal{C}(k)$ in total.

(d)

We assume two cases: either the source switch $s_{i,j}$ and the target switch $b_{m,n}$ share a pod ($i = m$) or they don't ($i \neq m$).

If $i = m$, the shortest path is necessarily going through one of the $\frac{k}{2}$ top-routers $s_{i,j}$ is connected to

$$s_{i,j} \to t_{i,x} \to b_{i,n}, \quad 1 \leq x \leq \frac{k}{2}$$

Each of the $\frac{k}{2}$ total paths has distance 2 and is disjoint. As only $\frac{k}{2} - 1$ links may fail at most, there will always be a shortest path left.

If $i \neq m$, the source is still connected to $\frac{k}{2}$ top-switches, which in turn each are connected to $\frac{k}{2}$ different backbone switches. From there, there only exists one possible shortest path: going to the top-switch of the target pod and from there to the target bottom-switch.

In total, there are $\frac{k^2}{4}$ shortest paths and because $\frac{k^2}{4} > \frac{k}{2} - 1$, there will always be a shortest path remaining after $\frac{k}{2} - 1$ links fail.

Exercise 2

(a)

In the following log all MAC addresses are marked yellow.

<MINTED>

(b)

```
mininet> sh ovs-appctl fdb/show s1
port  VLAN  MAC               Age
1     0  da:8f:53:5a:66:53   90
2     0  4a:a6:82:75:bb:0d   82
3     0  ba:4e:b6:40:6f:3d   69
```

The first column lists the physical ports of the switch. Each physical port has a number. The second column specifies in which VLAN the connected host is. The third lists the MAC address of the connected device, that is connected to the corresponding physical port. Age lists the last time in seconds the switch received a packet from. This is used to remove disconnected devices from the forwarding table later on.

(c)

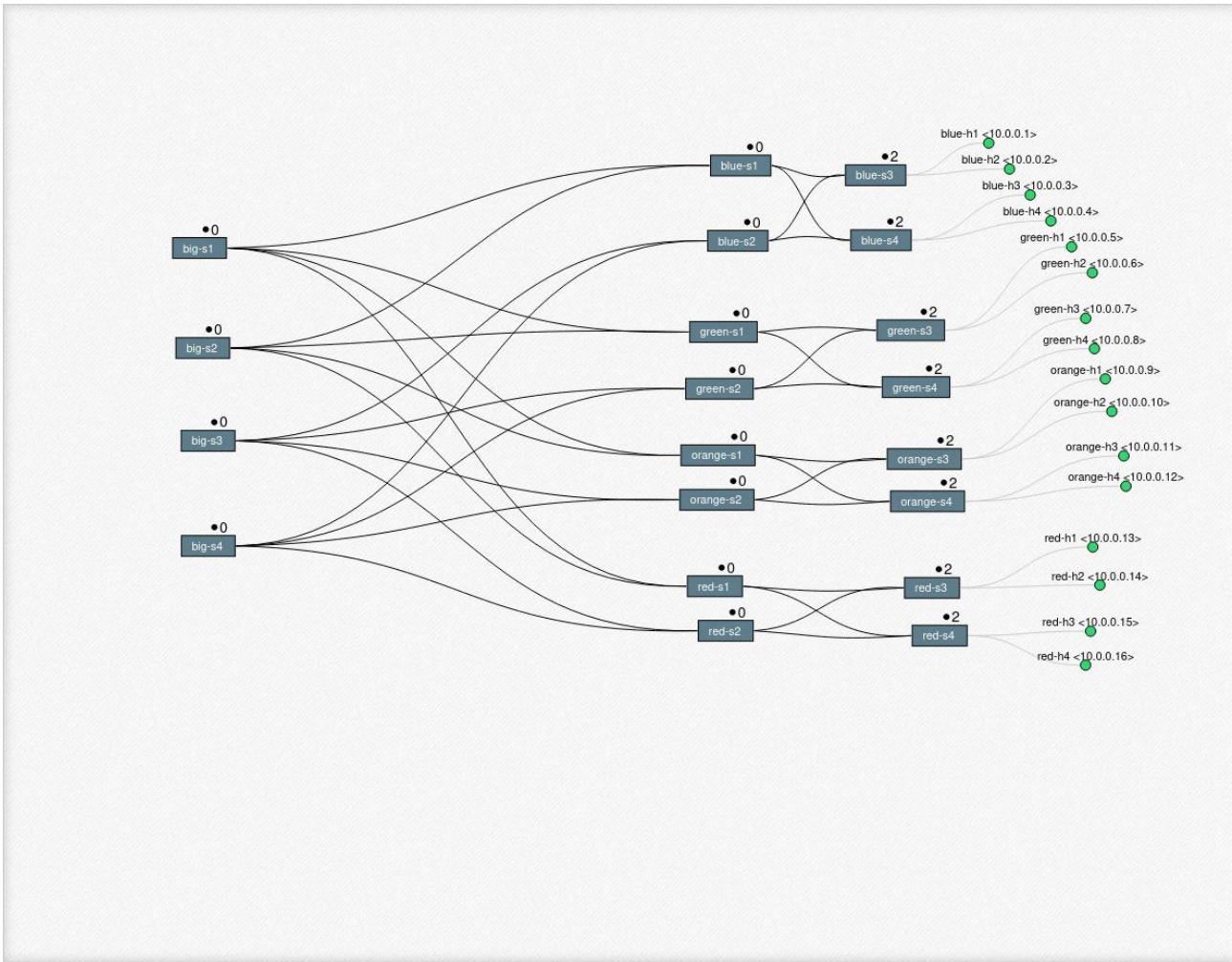


Figure 1: Fat Tree Topology on Mininet Topology Visualizer